

Capstone Project: Christmas Season Sales Analysis

What caused the 2020–2021 decline in Christmas revenue and profit, and what the team should examine next

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Data source: FP20 Analytics Challenge 12 (database `fp20c12`, table `dbo.xmas_sales`)

Seasons compared: 2019–2020 (baseline), 2020–2021 (decline), 2021–2022 (recovery)

Tool: Microsoft SQL Server (T-SQL)

Source files: `xmas.sql` (all queries) and `xmas_results.txt` (their output)

Executive Summary

Christmas-season revenue fell from \$7.18M in 2019–2020 to \$6.95M in 2020–2021, a drop of \$227,000 (-3.2%). Profit fell by the same percentage (-\$177,000), and the profit margin stayed at 76.9%.

The number of sales records did not fall (+1.0%), and the price paid per item barely moved (-0.4%). What changed was the number of items in each sales record, which fell 3.7%. The loss was concentrated in adult customers (18 and over), whose revenue fell \$353,000 (-15.2%) across all three sales channels, and in clothing and wearable tech. Growth in children's products offset part of it. Costs and product mix were tested and ruled out as causes of the profit decline.

The 2021–2022 season recovered \$135,000 of the lost revenue but ended 1.3% below the baseline. In-store sales recovered fully, while the Xmas Market and online channels both declined that season.

The data shows what changed and where, but not why adults bought less. Section 07 lists the follow-up analyses that would answer that and the decision each one supports.

Key figures, decline season compared with baseline

Measure	Baseline 2019–2020	Decline 2020–2021	Change
Revenue	\$7.18M	\$6.95M	-3.2%
Profit	\$5.52M	\$5.35M	-3.2%
Profit margin	76.93%	76.89%	-0.04 points
Sales records	9,924	10,023	+1.0%
Average value per record	\$723.41	\$693.62	-4.1%
Units per record	4.30	4.14	-3.7%

01. Background and Objective

The dataset records Christmas-season sales (November, December and January) for a supermarket operating in ten European countries. A first look at revenue by season shows steady growth until 2020–2021, when revenue and profit both fell.

This report has three objectives:

- Identify what kind of change caused the decline, and which customers, channels, products and places it came from.
- Establish whether profit fell for the same reason as revenue or for a separate one.
- Recommend the follow-up analyses the team should run before deciding how to respond.

02. Data and Technical Approach

The dataset

- **Size:** 41,163 rows and 19 columns, dated 1 January 2018 to 28 January 2022. All rows fall in November, December or January.
- **Contents:** each row holds a date and time, the customer's age range and gender, country and city, sales channel, payment method, product, quantity, unit price, cost, total sales and profit.
- **Coverage:** 10 countries, 36 cities, 3 channels (in-store, Xmas Market, online), 3 age groups, 5 payment methods, 13 product categories and 18 products.
- **What one row means:** the table has no order ID. A row is one product line (one customer buying some quantity of one product), so the report counts "sales records", not transactions or customers.
- **Rows analysed:** January 2018 is excluded because its season has no November or December data. That leaves four complete seasons (39,957 rows), of which the three compared seasons hold 29,981.

SQL techniques used

The script is written in T-SQL. Preparation logic is written once in two views, and every analytical query is built from them using the techniques below.

Technique	Where it is used	Purpose
Views (CREATE OR ALTER VIEW , batches separated by GO)	Section 0.5	Clean and label the data once, so all queries share the same definition of a season
Common table expressions (WITH ... AS), chained	Every query	Split a query into named steps: total, derive averages, then compare
Aggregation (GROUP BY , SUM , COUNT , AVG , HAVING)	Throughout	Totals per season and segment; HAVING finds duplicates and tests month coverage
Self-join on the previous season	Questions 2 and 3, Tasks 2 and 5	Place each season beside the one before it to calculate change
FULL OUTER JOIN with COALESCE and ISNULL	Tasks 3, 4 and 5	Put seasons side by side without dropping a segment that is missing from one season
NULLIF on every denominator	Every ratio	Return an empty result instead of a divide-by-zero error
CASE and date functions (EOMONTH , DATEDIFF , DATEADD , DATEPART)	Section 0.5, Question 10	Assign January to the season that began in November; derive weekday and hour

Technique	Where it is used	Purpose
UNION ALL	Sections 0 and 0.5	Build the season lookup table; stack the negative-value checks
INFORMATION_SCHEMA.COLUMNS	Section 0	List the table's columns and data types

Two conventions apply throughout: percentages are stored as proportions (0.032 means 3.2%), and margins are calculated from season totals, not by averaging row-level margins.

03. Analysis Approach

The analysis is divided into stages because each stage answers one question, and its answer decides what the next stage needs to ask. The headline number alone (revenue down 3.2%) cannot be acted on, so each stage narrows it down further.

Stage	Question	Method
Section 0	Can the data be trusted?	Check for missing values, duplicate rows, sales that do not equal cost plus profit, and negative values
Section 0.5 (Task 1)	What data is analysed?	Build a cleaned view (complete seasons only, each row tagged with its season) and a lookup naming the baseline, decline and recovery seasons
Questions 1 to 3	What happened, and how large was it?	Total revenue, quantity, cost and profit per season; compare each season with the previous one
Task 2	What kind of change was it?	Split revenue into number of records, units per record and revenue per unit, and see which one moved
Task 3 (Questions 4 to 10)	Where did it happen?	Compare seasons one category at a time: channel, place, age, gender, payment method, product and time
Task 4	Do those findings overlap?	Combine two categories at once, and compare units sold with price for each product
Task 5	Did profit fall for the same reason?	Compare cost per unit and profit margin overall, by category and by channel
Task 6	Do the parts add up to the whole?	Add channel revenue back up and compare it with the season total

The seasons were labelled after Question 1 showed that 2020–2021 was the only season in which revenue fell. The season before it became the baseline and the season after it the recovery.

04. Findings

4.1 Data quality (Section 0)

All checks passed: no missing values in the 15 columns the analysis depends on, no duplicate rows, no rows where sales differ from cost plus profit, and no negative values. No cleaning was needed beyond removing the incomplete 2017–2018 season.

4.2 What happened (Questions 1 to 3)

Revenue grew 1.5% into 2019–2020, fell 3.2% in 2020–2021 and grew 1.9% in 2021–2022. Quantity and profit followed the same path. The decline interrupted an upward trend, and the recovery did not return revenue to its previous level.

Table 1. Sales by Christmas season

Season	Revenue	Change	Units sold	Profit
2018–2019	\$7.07M	n/a	42,600	\$5.44M
2019–2020 (baseline)	\$7.18M	+1.5%	42,700	\$5.52M
2020–2021 (decline)	\$6.95M	-3.2%	41,500	\$5.35M
2021–2022 (recovery)	\$7.09M	+1.9%	42,400	\$5.45M

4.3 What kind of change it was (Task 2)

Revenue equals the number of sales records multiplied by the average value of a record, and that average value equals units per record multiplied by revenue per unit. Measuring each part separately shows which one caused the decline.

Table 2. Revenue broken into its parts

Season	Sales records	Average value per record	Units per record	Revenue per unit
Baseline	9,924	\$723.41	4.30	\$168.15
Decline	10,023 (+1.0%)	\$693.62 (-4.1%)	4.14 (-3.7%)	\$167.44 (-0.4%)
Recovery	10,034 (+0.1%)	\$706.25 (+1.8%)	4.22 (+1.9%)	\$167.23 (-0.1%)

The number of records rose slightly and the price per unit was almost unchanged. The decline came from fewer units in each record. Customers kept buying, at the same prices, but bought less each time.

4.4 Where it happened (Task 3)

Age group. This is the clearest split in the data. Adults lost more revenue than the company as a whole, and children's products grew.

Table 3. Revenue by customer age group, decline compared with baseline

Age group	Baseline	Decline	Revenue change	% change	Average value per record
18 and over	\$2.32M	\$1.97M	-\$352,695	-15.2%	-11.7%
12 to 17	\$2.14M	\$2.14M	-\$4,683	-0.2%	-5.7%
1 to 11	\$2.71M	\$2.85M	+\$130,363	+4.8%	+3.5%

Channel. In-store lost the most revenue in the decline season and then recovered fully. The Xmas Market declined in both seasons. Online grew during the decline and then fell sharply.

Table 4. Revenue by sales channel across the three seasons

Channel	Baseline	Decline	Recovery	Decline vs baseline	Recovery vs decline
In-store	\$4.60M	\$4.40M	\$4.76M	-4.3%	+8.3%
Xmas Market	\$1.29M	\$1.23M	\$1.19M	-4.8%	-3.4%
Online	\$1.29M	\$1.32M	\$1.13M	+2.4%	-14.2%

The remaining splits are summarised below.

- **Product category:** clothing lost \$150,000 (-16.8%), wearable tech \$75,000 (-26.2%) and kitchen appliances \$53,000 (-18.5%). Puzzles (+20.5%), board games (+9.0%) and building sets (+6.2%) grew.
- **Country:** the United Kingdom lost \$141,000 (-14.7%), Poland \$97,000 (-18.1%) and Germany \$74,000 (-7.5%). London alone lost \$101,000 (-35.6%). France (+14.5%), Belgium (+9.9%) and Portugal (+8.2%) grew.
- **Gender:** revenue from female customers fell 6.0% (\$219,000); revenue from male customers was flat (-0.2%).
- **Payment method:** PayPal (-10.6%), debit card (-10.8%) and credit card (-4.6%) fell; cash grew 2.3%.
- **Month:** December revenue fell 6.1% and November 3.0%, while January grew 4.7%.

4.5 How the findings overlap (Task 4)

Looking at one category at a time can count the same lost sales several times. Task 4 combines categories to check.

- **Adults bought less in every channel:** in-store -\$180,000 (-13.7%), Xmas Market -\$101,000 (-19.5%) and online -\$72,000 (-14.5%). The adult decline is not a problem with one channel.
- **Clothing fell in the physical channels:** Xmas Market clothing lost \$83,000 (-38.6%) and in-store clothing \$63,000 (-12.7%), while online clothing was nearly flat (-2.2%).
- **The payment pattern is not uniform:** card and PayPal revenue fell in-store, but credit-card revenue grew online (+5.6%). The payment finding in 4.4 mostly reflects where adults shop.

The products that lost the most revenue sold far fewer units, while their prices moved much less. In each case the fall in units was at least twice the size of the price change.

Table 5. Units and price for the five products with the largest revenue loss

Product	Units, baseline	Units, decline	Units change	Average unit price change	Revenue change
The North Face Jacket	1,981	1,589	-19.8%	-1.3%	-\$79,000
Fitbit Inspire	1,645	1,396	-15.1%	-6.4%	-\$75,000
Nike Running Shoes	1,677	1,390	-17.1%	-3.4%	-\$69,000
Dolce Gusto Coffee Machine	1,678	1,412	-15.9%	-4.9%	-\$53,000
Ray-Ban Sunglasses	1,979	1,597	-19.3%	+5.0%	-\$49,000

4.6 Why profit fell (Task 5)

Profit fell \$177,000 (-3.2%), the same percentage as revenue. Cost per unit was \$38.79 in the baseline and \$38.69 in the decline, and the margin moved by 0.04 points. Every product category and every channel has a margin of about 77% in both seasons, so a shift between them could not have changed overall profitability.

Profit therefore fell only because less was sold, and in the same places as revenue: clothing lost \$116,000 of profit, wearable tech \$58,000 and the in-store channel \$152,000.

4.7 Reliability check (Task 6)

For each of the three seasons, the revenue of the three channels adds up exactly to the season total (a gap of \$0.00). No revenue sits outside the channel breakdown.

05. Discussion

One explanation, not several

The findings describe a single change seen from different angles. Sales records got smaller because they contained fewer units. That happened mainly among adults, in every channel, and in the products adults typically buy: clothing, wearable tech and kitchen appliances. Profit fell because sales fell.

What was ruled out

Each alternative explanation was tested directly.

Possible explanation	Evidence	Conclusion
Fewer purchases were made	Sales records rose 1.0%	Ruled out
Prices were lower	Revenue per unit fell only 0.4%	Ruled out
Costs rose	Cost per unit fell slightly, from \$38.79 to \$38.69	Ruled out
Customers moved to lower-margin products or channels	Margins are about 77% in every category and channel	Ruled out
One channel failed	Adult revenue fell 14% to 20% in all three channels	Ruled out
Fewer units per record	Units per record fell 3.7%; top losing products sold 15% to 20% fewer units	Supported

Scale

The adult loss (\$353,000) is larger than the company's total loss (\$227,000). The difference is growth elsewhere, mainly children's products (+\$130,000). The business did not shrink evenly: one customer group fell sharply while another grew.

The recovery is incomplete and uneven

Revenue in 2021–2022 was still 1.3% below the baseline. Adult revenue recovered to \$2.20M but remained about 5% below its baseline, and revenue from the 1-to-11 age group fell back to about

3% below its baseline. The in-store channel exceeded its baseline, while the Xmas Market fell for a second season and online dropped 14.2%. The overall recovery figure hides these opposite movements.

Limits of the analysis

- **The data does not explain motive.** It shows that adults bought fewer items, not why. Possible reasons such as tighter household budgets, stock shortages or outside events that season cannot be tested with this table.
- **A record is not a transaction.** Without an order ID, the report cannot tell whether customers made fewer shopping trips or bought fewer product lines per trip.
- **Margins are gross margins.** The cost column appears to cover the goods only, so the 77% margin does not include staff, rent, delivery or marketing.
- **One declining season.** The decline is a single season of 3.2%, so small segments (for example, one product in one channel) should be read with caution.

06. Summary of Key Findings

1. Revenue and profit both fell 3.2% in 2020–2021; the margin stayed at 76.9%.
2. The cause was fewer units per sales record (-3.7%), not fewer records (+1.0%) or lower prices (-0.4%).
3. Adults account for the decline: -\$353,000 (-15.2%), in all three channels.
4. Clothing and wearable tech lost the most, through lower unit sales at similar prices.
5. Costs and product mix did not contribute to the profit decline.
6. The 2021–2022 recovery was partial, and the Xmas Market and online channels declined in it.

07. Recommendations for Further Analysis

The findings identify where the decline sits but not its cause, so the recommendations below are investigations. Each one states what is still unknown, what to check and the decision the answer would support.

1. Audit what adults stopped buying

- **What we know:** Adults lost \$353,000, and clothing and wearable tech lost the most, but the script does not yet cross age group with product.
- **What to check:** Extend Task 4 with an age group by product category comparison, showing units per record and revenue per unit for baseline, decline and recovery.
- **Decision it supports:** Whether to respond with a promotion, a change in the product range or a lower-priced tier aimed at adults.

2. Test whether clothing and wearable tech were a supply problem

- **What we know:** Unit sales of five products fell 15% to 20% at similar prices. Lower demand and running out of stock would both look like this.
- **What to check:** Chart daily units for those products in both seasons and look for gaps or early cut-offs; compare with inventory records if the team has them.

- **Decision it supports:** Whether the fix belongs to purchasing and stock planning or to marketing.

3. Review the Xmas Market channel on its own

- **What we know:** It is the only channel that declined in both seasons (-4.8%, then -3.4%), and its clothing revenue fell 38.6%.
- **What to check:** Repeat the age, product and country comparisons for this channel only, across all three seasons.
- **Decision it supports:** Whether the Xmas Market needs its own plan or will recover once the adult decline is addressed.

4. Explain the online drop in the recovery season

- **What we know:** Online grew 2.4% during the decline and then fell 14.2%, the opposite of the in-store pattern.
- **What to check:** Break online revenue in 2021–2022 down by month, product category and country, and compare the timing with any changes to the website, delivery or pricing.
- **Decision it supports:** Whether online needs a service fix, or whether customers simply returned to stores.

5. Check whether the country results are the same pattern or a local one

- **What we know:** The United Kingdom and Poland lost the most, and London alone fell 35.6%, while France and Belgium grew.
- **What to check:** Compare age group by country. If adults fell everywhere and the United Kingdom simply has more adult revenue, the cause is the same; if not, something local happened.
- **Decision it supports:** Whether to run one company-wide response or country-specific ones.

6. Follow each segment through to the recovery season

- **What we know:** Most comparisons in Tasks 4 and 5 stop at the decline season, and the overall recovery hides segments still below baseline.
- **What to check:** Add recovery-versus-baseline columns to the Task 4 and Task 5 queries.
- **Decision it supports:** Which segments still need attention and which have already corrected themselves.

7. Close the two data gaps

- **What we know:** The cost column's definition is unconfirmed and there is no order ID.
- **What to check:** Confirm with the data owner what cost includes. If an order ID becomes available, replace the row counts marked -- **GRAIN** in the script with distinct order counts and rerun Task 2.
- **Decision it supports:** Whether margins can be used to compare channels, and whether the response should aim at more visits or at larger baskets.

08. Data Quality and Assumptions

- **Missing values:** 0 in each of the 15 columns the analysis depends on.
- **Duplicate rows:** 0, comparing all 19 columns.

- **Sales = cost + profit:** holds for every row (tolerance of \$0.01).
- **Negative values:** 0 in total sales, quantity, cost, profit and unit price.
- **Season definition:** November and December plus the following January. A season is included only if all three months have data reaching the first and last week of the month.
- **Dates:** every month in the table ends on the 28th, so the last days of each month are not recorded. This applies equally to all seasons and does not affect the comparisons.
- **Counts:** "sales records" are rows of the table (product lines), not confirmed transactions.
- **Verification:** the full script was last run on 7 October 2026 against a local SQL Server 2022 restore of the database. All 34 result sets returned without errors or warnings.

Appendix: Where Each Result Comes From

Report section	Part of <code>xmas.sql</code>
4.1 Data quality	Section 0 (checks 0a to 0g)
4.2 What happened	Questions 1, 2 and 3
4.3 What kind of change	Task 2
4.4 Where it happened	Task 3: Questions 4 (channel), 6 (place), 7 (age, gender, payment), 9 (product) and 10 (time)
4.5 How the findings overlap	Task 4: parts 4a to 4d
4.6 Why profit fell	Task 5: parts 5a to 5c
4.7 Reliability check	Task 6